Semantic-Driven Hierarchical Fusion With Adaptive Reasoning for Weak-Information Target Change Detection in Remote Sensing Images

Xiang Li, Biao Wang, *Member, IEEE*, Zhenghao Jiang, Peng Zhang, and Yu Meng

***Abstract*—The semantic change detection (SCD) plays a critical role in the intelligent information extraction from remote sensing (RS) images. However, its performance in detecting weak-information targets, such as small-sized ground objects, is still constrained by the semantic representation ambiguity, information loss, and complex background interference. This article proposes a semantic guidance inference network (SGI-Net) that establishes a collaborative optimization mechanism through the semantic-driven feature coupling, adaptive reasoning, and attention guidance to overcome these challenges. Specifically, we design a hierarchical feature coupling module (HFCM). This module enhances the geometric detail representation for weak-information targets through the cross-layer feature aggregation and dual-path decoupling. It generates discriminative semantic priors, establishing geometric-semantic correlations across multilevel features. Furthermore, a multidimensional adaptive inference module (MAIM) is developed to achieve the adaptive feature aggregation of local–global patterns via nonoverlapping patch partitioning, preserving spatial continuity of weak-information targets through the multiscale feature fusion. In addition, the change-guided attention (CGA) module dynamically adjusts the channelwise and spatial attention weights based on the prior knowledge, amplifying cross-hierarchical responses to weak change signals. Experimental results on the SECOND and SJH_SCD datasets demonstrate that SGI-Net achieves values of 73.86% and 77.43%, respectively, surpassing state-of-the-art (SOTA) methods by 0.73%–1.37% in mIoU and 1.26%–1.77% in separated Kappa coefficient (SeK). The visualization analysis further confirms that SGI-Net effectively enhances the feature responses of weak-information targets in complex object scenes, improving the discrimination accuracy of fine-grained change boundaries. These results provide the reliable technical support for high-precision RS change detection (CD).**

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I. Introduction

SEMANTIC change detection (SCD) aims to identify and quantify semantic changes in land cover across multitemporal remote sensing (RS) images [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [60]. By analyzing images acquired over different time periods, SCD can reveal trends in environmental changes, thereby providing the accurate decision support for resource management [14], urban planning [15], and disaster response [16]. With the increasing availability of high-resolution satellite and uncrewed aerial vehicle (UAV) images, a key challenge in SCD lies in advancing semantic interpretation from passive recognition to the active cognition [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [60]. Therefore, enhancing SCD's capability to incorporate expert knowledge and optimize semantic reasoning is of significant research importance and practical value. This advancement will facilitate the transition of RS applications from merely detecting physical changes to achieving a deeper understanding of environmental evolution.

Researchers have widely adopted deep learning techniques for change detection (CD) in RS images (RSIs) [17], [18], [19], [61], [62], [63]. Compared with traditional methods that rely on handcrafted features, CD operators, or threshold analysis to identify changed areas [20], [21], deep learning models can leverage the statistical consistency between multitemporal images to automatically learn and extract nonlinear features, thereby improving the robustness and adaptability of CD. Particularly in complex scenarios, models such as convolutional neural networks (CNNs) [22], [23] and generative adversarial networks (GANs) [24] have demonstrated the superior feature representation capabilities. For instance, the joint sparse representation (JSR) [25] and CNN-based spatial structure extraction methods [26] have been extensively used in CD tasks. However, while these methods effectively identify the shape and location of changed areas in binary CD (BCD), they fail to further distinguish change types, thereby limiting their applicability in refined analysis. Consequently, the SCD has emerged as a natural progression to bridge this gap. SCD not only identifies "where" changes occur but also analyzes "what" types of changes have taken place [1], [2], [3], [4], [5],

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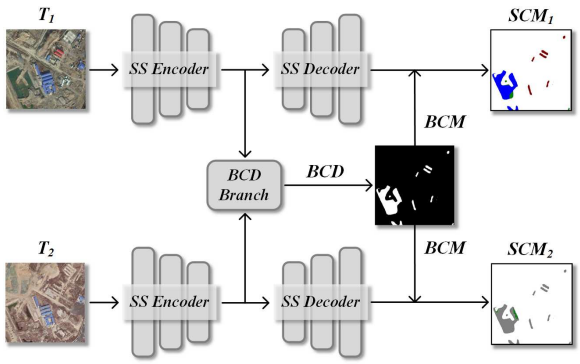


Fig. 1. Illustration of the SCD task.

[6], [7], [8], [9], [10], [11], [12], [13], [60]. As shown in Fig. 1, the standard output of SCD consists of two key components: 1) bitemporal semantic change maps (SCM_1 and SCM_2), which represent the semantic information of the changed regions at the two time points, respectively, and 2) a binary change map (BCD), which indicates the locations of the changes. The final semantic change information is generated by applying BCD as a mask to the SCM results, assigning a ($class_i$ and $class_j$) label [e.g., (building and water)] to each changed pixel. This output paradigm is widely adopted in mainstream SCD methods, as it not only preserves the independent semantic information of both time points—supporting land-use analysis—but also enables the straightforward extraction of "from–to" change types via simple masking operations.

Existing SCD methods primarily leverage deep neural networks to learn feature variations from multitemporal images, integrating multiscale feature fusion strategies to enhance the detection accuracy [27], [28], [29], [30]. CNN-based encoder–decoder structures [22], [23] and Transformer-based long-range dependency modeling [31], [32] have achieved remarkable progress in large-scale scene CD tasks, such as urban expansion and forest cover change, demonstrating strong applicability. However, these methods still face significant limitations, constrained by imbalanced sample distributions and sensitivity to feature scales, deep learning models often focus excessively on areas with significant changes. This leads to insufficient representation of small targets, making it challenging to effectively capture features of weak-information targets, such as tiny objects [35], [36], [37]. Particularly in fine-grained CD involving high-resolution imagery, issues such as missed detections and false detections of weak-information targets become key challenges that limit detection accuracy.

In this study, weak-information targets primarily encompass size-driven objects (e.g., small buildings and narrow roads), while also extending to targets with low-change magnitude (e.g., gradual vegetation shifts) or strong background interference (e.g., roads obscured by tree canopy) [38], [39], [40], [41], [42], [43], [44], [45]. Although these targets occupy only a few pixels in the images, their subtle spatial changes often carry critical geographic semantic information. This information is essential for decision-making in applications, such as urban planning and disaster assessment [14], [15], [16], [43], [44], [45].

To address the challenges of detecting weak-information targets in RSIs, researchers have proposed various strategies in recent years to improve detection performance. The primary approaches include the following.

1) *Multibranch Feature Extraction:* Xu et al. [39] employed multibranch network architectures to capture features at different scales and hierarchical levels. In addition, depthwise separable convolutions (DSCs) are used to extract spatial features, enhancing the localization capability for small targets.
2) *Coarse-to-Fine Detection Strategies:* Yao et al. [40] adopted a multistage detection framework, where the coarse stage leverages structured edge-based nonlocal self-similarity features to distinguish targets from the background, while the fine stage integrates local contrast priors to enhance the discriminability of small targets.
3) *Local Region Awareness and Multiscale Feature Fusion:* Zhang et al. [41] improved local region perception, integrated multiscale features and established global associations across channels and spatial domains to mitigate issues related to insufficient feature representation and background confusion for small targets.
4) *Sparse-View Network:* Lei and Liu [42] introduced a sparse-view attention mechanism, coupled with a sparse-view path aggregation network (SOLO-PAN), to enhance the model's focus on small targets. Although these methods have improved the detection performance of weak-information targets to some extent, critical challenges persist.
    a) *Limited Feature Representation Capability:* Due to their small size, weak-information targets in RS imagery often exhibit low brightness, blurred edges, or weak contrast, making them difficult to capture with existing detection methods. While multiscale feature extraction helps in target localization, further research is needed to enhance the fine-grained details of small targets.
    b) *Inherent Limitations of the Encoder–Decoder Structure:* Most existing methods rely on a downsampling–upsampling encoder–decoder framework, where small-target information is progressively lost in deep networks, making it challenging to fully recover fine details and affecting detection completeness.
    c) *Poor Adaptability to Low-Contrast Scenarios:* In complex backgrounds, weak-information targets often exhibit low contrast with their surroundings, making their features susceptible to environmental interference. This leads to difficulties in accurately distinguishing and extracting subtle changes using conventional detection methods.

Therefore, key challenges in CD tasks include further strengthening the feature representation of weak-information targets, optimizing detection frameworks to minimize the information loss, and improving model robustness in low-contrast environments.

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Based on the aforementioned research progress and limitations, this article proposes the semantic guidance inference network (SGI-Net), which focuses on extracting semantic evolution features of weak-information targets, such as small objects. The main contributions of this study are as follows.

1) To enhance the feature representation of weak-information targets, this study embeds a hierarchical feature coupling module (HFCM) at the end of the dual encoder. It aggregates cross-level features to construct geometric-semantic priors and extracts contextual features through multiscale sampling. A novel dual-path feature decoupling strategy is then applied to enable dynamic interaction between deep semantic features and shallow geometric details. This process generates more discriminative feature representations for weak-information targets.
2) To address the loss of weak-information targets during the downsampling stage, this study introduces a multidimensional adaptive inference module (MAIM) in the feature fusion phase. A parallel fusion architecture based on a nonoverlapping patch strategy is used to achieve collaborative optimization of local detail enhancement and global contextual constraints. Combined with dual-domain attention in both the channel and spatial dimensions, this approach enables adaptive feature aggregation, effectively preserving the spatial continuity of weak-information targets.
3) To effectively suppress complex background interference and focus the network on the change details of weak-information targets, this study uses HFCM to generate a change prior map. A change-guided attention (CGA) module is then applied to dynamically adjust attention weights, suppressing background noise while enhancing cross-level responses to weak change signals.

The remainder of this article is organized as follows. Section II details the SGI-Net architecture. Section III introduces the datasets and parameter settings. Section IV validates the method through comparative experiments and visual analysis. Finally, Section V concludes this article and discusses future research directions.

II. Methodology

The overall architecture of the proposed SGI-Net is illustrated in Fig. 2. SGI-Net adopts a dual-branch encoder and triple-branch decoder framework to efficiently extract and fuse change information and semantic features from multitemporal RSIs.

In the encoder stage, SGI-Net employs ResNet34 [46], [47] as the backbone and utilizes a Siamese weight-sharing structure [48] to capture multiscale features from bitemporal images. The extracted features F_0, F_1, and F_4 are sequentially fed into the HFCM to enhance the semantic representation and spatial detail. The aggregated features are then passed as prior information to the first MAIM to generate a change-guidance map. This map is further refined using the CGA module to focus on changes in weak-information targets, thereby providing guidance for subsequent feature fusion.

During the feature fusion stage, bitemporal features extracted by the dual-branch encoder are progressively fed into MAIM, where the change-guidance map facilitates feature fusion and enhances change discrimination. CD and semantic segmentation (SS) results for bitemporal RSIs are predicted by independent classifiers, enabling fine-grained SCD.

A. Hierarchical Feature Coupling Module

As a typical dense prediction task, SCD still faces challenges in effectively representing weak-information targets [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [38]. These targets inherently exhibit characteristics, such as low brightness, blurred boundaries, and low contrast, making them difficult for deep networks to capture under complex background interference, ultimately affecting the overall detection accuracy [39], [40], [41], [42], [43], [44], [45]. To enhance the network's focus on these critical regions and improve the detection capability of weak-information targets, introducing representative prior information before feature fusion is an effective strategy. This prior information can guide the subsequent feature fusion process, ensuring the network concentrates more on significant change regions [49], [50], [51], [52], [53].

Based on this, this study focuses on the detection of size-driven weak-information targets and proposes a strategy that effectively preserves critical details, addressing the limitations of existing methods. Current deep learning models are primarily reliant on high-level semantic features learned from images. However, such reliance on deep features alone often leads to the loss of fine details in weak-information targets, making them difficult to identify accurately. We argue that, in addition to rich semantic information, low-level features containing shape, edge, and other structural details are equally crucial. Traditional approaches typically use skip connections to directly transfer unrefined features, resulting in insufficient fusion of deep semantics and shallow details. This limitation is particularly pronounced in detecting weak-information targets, where precise geometric boundary delineation is critical. To address these challenges, the proposed SGI-Net incorporates an HFCM at the end of the dual-branch encoder. By leveraging a semantic-driven feature coupling mechanism, HFCM facilitates cross-hierarchy feature interaction, as illustrated in Fig. 3.

The HFCM constructs a multilevel perception system through cross-layer feature fusion, generating multidimensional prior information to guide subsequent processing. It receives three features from the encoder: the shallow layer F_0 (high-resolution detail features), the middle layer F_1 (spatial structure features), and the deep layer F_4 (high-level semantic features). $F_0 \in \mathbb{R}^{C_0 \times H_0 \times W_0}$ retains edge textures for localization, $F_1 \in \mathbb{R}^{C_1 \times H_1 \times W_1}$ balances detail and structure to improve foreground–background separation, and $F_4 \in \mathbb{R}^{C_4 \times H_4 \times W_4}$ encodes high-level semantic context essential for scene interpretation; yet, its spatial degradation limits direct identification of weak-information targets. HFCM overcomes this by coupling F_4 with F_0 (retaining edge textures) and F_1 (preserving structural details) to establish cross-hierarchy correlations. Although spatial compression in deep layers (e.g.,

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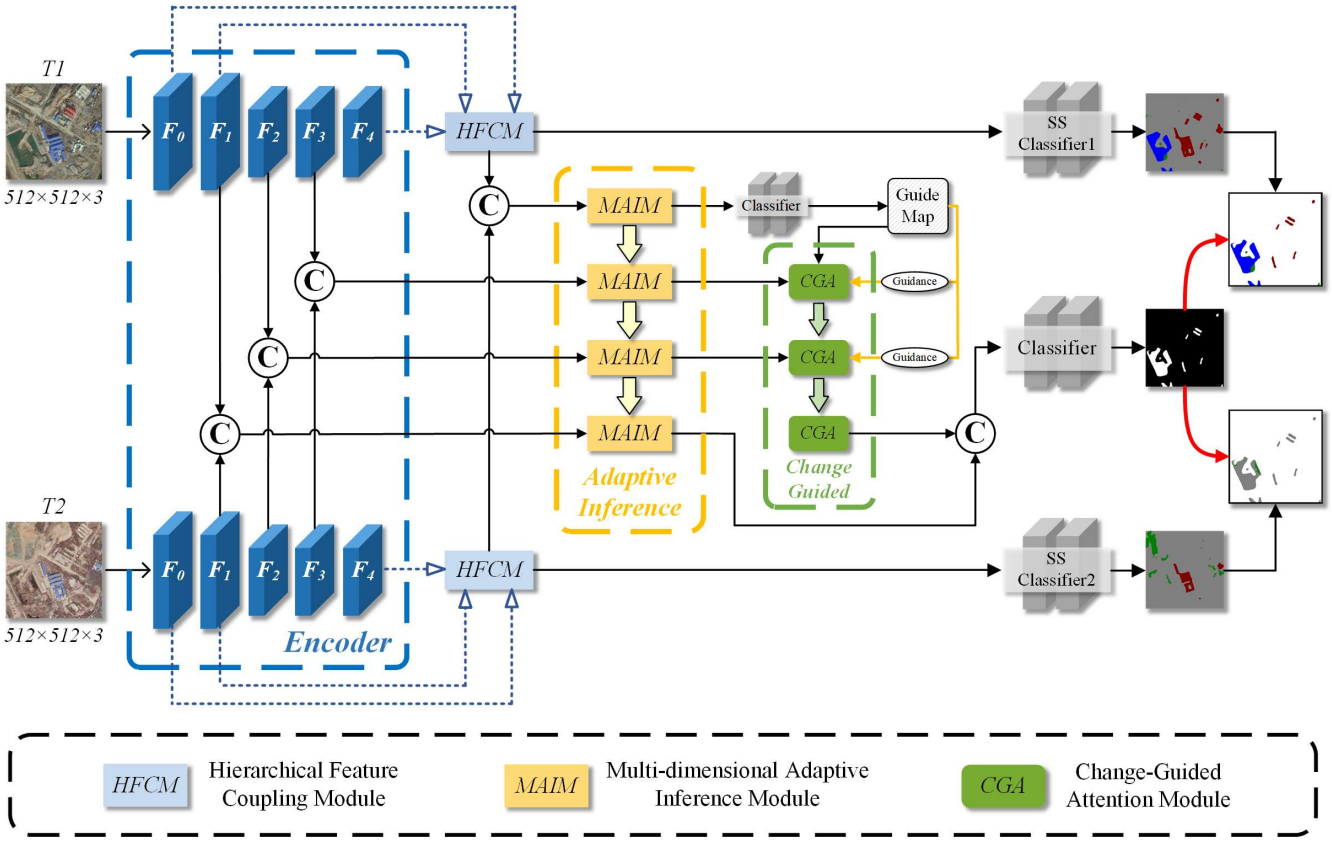


Fig. 2. Overall architecture of SGI-Net, comprising a dual-branch encoder, HFCM, MAIM, and CGA.

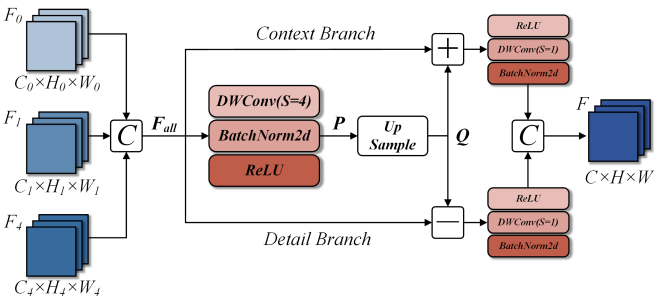


Fig. 3. Proposed HFCM. Processes multilevel features (F_0, F_1, and F_4) via a dual-path mechanism (addition for context and subtraction for local differences) to generate geometric-semantic priors.

downsampling) may degrade geometric details in F_4, particularly for small objects, HFCM compensates by explicitly fusing F_0 and F_1, which preserve crucial high-resolution spatial information. By fusing these features and applying a dual-path residual mechanism, HFCM enhances weak-target representation and mitigates the loss of global context in low-level features, thereby improving CD performance [55], [56]. Section IV further evaluates the benefits of this hierarchical fusion.

To align the spatial resolutions of F_0, F_1, and F_4 before concatenation, we downsample features hierarchically. Specifically, F_0 is pooled to match F_1's resolution, yielding $\mathbb{R}^{C_0\times H_1\times W_1}$. This pooled F_0 is then concatenated with F_1 along channels, producing $\mathbb{R}^{(C_0+C_1)\times H_1\times W_1}$. The result is downsampled to F_4's resolution and added to F_4

$$F_{\text{all}} = \text{Downsample}\left(\text{Concat}\left(\text{Pool}\left(F_0\right), F_1\right)\right) + F_4. \quad (1)$$

Thus, $F_{\text{all}} \in \mathbb{R}^{C_{\text{all}}\times H_{all}\times W_{\text{all}}}$, where H_4 and W_4 denote F_4's spatial dimensions. This reduces computation while preserving weak-target details.

Then, F_{all} is processed using a 5×5 DSC to obtain a downsampled feature map $P \in \mathbb{R}^{C_{all}\times H_{\text{all}}/2\times W_{\text{all}}/2}$, which is then upsampled back to the same spatial resolution as F_{all}, resulting in feature Q

$$Q = \text{Upsample}\left(\text{DWConv}_{5\times 5}\left(F_{\text{all}}\right)\right). \quad (2)$$

The choice of DSC is motivated by its efficiency in parameter reduction and local feature enhancement, crucial for preserving weak-target details. By decoupling spatial and channel processing, DSC maintains geometric sensitivity while reducing redundancy.

Here, enabling a residual-based dual path: one branch computes $F_{\text{all}} - Q$ to enhance fine details and the other uses $F_{\text{all}} + Q$ to capture contextual information. To explicitly balance fine-detail enhancement and contextual consistency, the dual-path design in HFCM operates by separating frequency components through addition and subtraction. The subtractive (high frequency) path computes $F_{\text{all}} - Q$ to extract residuals that emphasize edges, textures, and other local geometric cues, thereby amplifying the boundaries of weak-information targets. In contrast, the additive (low frequency) path computes

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$F_{\text{all}} + Q$ to aggregate low-frequency components, which reinforces global shape and semantic coherence while suppressing high-frequency noise. These two complementary streams are then concatenated and passed through a projection convolutional layer, enabling the dynamic fusion of semantic context with fine-grained structural details. Both paths pass through 3×3 depthwise convolutions with batch normalization, are concatenated

$$F_{\text{detail}} = \text{BN}\left(\text{DWConv}_{3\times3}\left(\text{ReLU}\left(F_{\text{all}} - Q\right)\right)\right) \tag{3}$$

$$F_{\text{context}} = \text{BN}\left(\text{DWConv}_{3\times3}\left(\text{ReLU}\left(F_{\text{all}} + Q\right)\right)\right). \tag{4}$$

Here, ReLU and BN denote the rectified linear unit and batch normalization, respectively. Also, finally recalibrated by a 1×1 convolution to produce the output F with rich contextual and spatial information

$$F = \text{ReLU}\left(\text{Conv}\left(\text{Concat}\left(F_{\text{detail}}, F_{\text{context}}\right)\right)\right). \tag{5}$$

B. Multidimensional Adaptive Inference Module

MAIM serves as a core component for the proposed network, aiming to mitigate the issue of weak-information targets being progressively lost and difficult to recover in traditional encoder–decoder architectures. To address the challenges posed by complex background interference and the loss of fine details during the encoding stage, MAIM employs a nonoverlapping block-based multiscale parallel fusion architecture to optimize computational efficiency while maintaining spatial coherence. The intrapatch averaging acts as a low-pass filter that preserves statistical consistency (e.g., mean intensity) of local regions, with multiscale fusion restoring continuity across patches. This design enables adaptive calibration of both local and global features through dynamic weight allocation, thereby enhancing feature representation capability.

Specifically, MAIM is designed with three parallel branches (as shown in Fig. 4): local branch, global branch, and skip branch. First, the two feature tensors $F_1 \in \mathbb{R}^{C_1\times H_1\times W_1}$ and $F_2 \in \mathbb{R}^{C_2\times H_2\times W_2}$ from the same hierarchical level but different temporal instances are aggregated via elementwise addition to obtain $F_m \in \mathbb{R}^{C_m\times H_m\times W_m}$. Next, different patch sizes (e.g., 2×2 and 4×4) are used to distinguish between the local and global branches. The rationale behind this distinction lies in the fact that patch size directly determines the receptive field for feature extraction, thereby influencing the model's sensitivity to local details and global context. A 2×2 patch contains information from only four adjacent pixels, capturing highly localized features such as edges, textures, or fine structural details. In contrast, a 4×4 patch covers 16 adjacent pixels, providing a fourfold increase in the receptive field. This enables the model to integrate broader contextual information, such as the relative position of objects, overall contours, and relationships with the background. In the experimental results presented in Section IV, this study further investigates the rationale behind the selection of p, as well as the impact of patch size on performance.

This module leverages a nonoverlapping patch mechanism to integrate local details and global context, enhancing the detection of weak-information targets. Unlike atrous spatial pyramid pooling (ASPP) [54], which relies on dilated convolutions and suffers spatial distortion from downsampling, this approach more accurately highlights subtle changes in complex backgrounds. The input feature $F_m \in \mathbb{R}^{C_m\times H_m\times W_m}$ is first fused by a 3×3 convolution on the sum of two feature maps

$$F_m = \text{Conv}_{3\times3}\left(F_1 + F_2\right). \tag{6}$$

Then, F_m is split into nonoverlapping patches of size $p \times p$

$$F_{p=2,4} = \text{Flatten}\left(F_m \rightarrow (p \times p)\,\text{patches}\right). \tag{7}$$

Here, Flatten represents the unfolding operation.

Each patch is averaged across channels to retain spatial structure and remove channel redundancy. The resulting vector is passed through two MLP layers with LayerNorm

$$F_{p=2,4} = \text{MLP}_2\left(\text{LayerNorm}\left(\text{MLP}_1\left(\text{Mean}\left(F_{p=2,4}\right)\right)\right)\right). \tag{8}$$

Here, Mean denotes the mean operation and MLP and LayerNorm correspond to a multilayer perceptron and layer normalization, respectively. This nonoverlapping patch division with intrapatch averaging reduces redundancy by compressing correlated local features (e.g., $p \times p \times C \rightarrow 1 \times 1 \times C$) while suppressing high-frequency noise. Spatial continuity is preserved through implicit positional encoding in MLP layers and skip-connection compensation in MAIM.

A spatial Softmax then produces attention weights, which are multiplied elementwise with the patch to emphasize salient regions

$$F_{p=2,4} = \text{Softmax}\left(F_{p=2,4}\right) \odot F_{p=2,4}. \tag{9}$$

Here, Softmax performs probability normalization and $\odot$ signifies elementwise multiplication.

Finally, the original feature F_m and the two patch-level features $F_{p=2}$ and $F_{p=4}$ are concatenated and reweighted via a channel–spatial attention mechanism, dynamically fusing local detail with global context

$$F = \text{Concat}\left(F_m, F_{p=2}, F_{p=4}\right). \tag{10}$$

This design compresses redundancy while enhancing response to weak-information targets and suppressing background interference.

It is important to note that MAIM processes only features from the same hierarchical level but at different time steps. In other words, handling features at a single level alone may lead to significant loss of relevant information. Since our objective is to detect weak-information targets, focusing solely on one feature level is insufficient. Shallow network layers preserve fine-grained spatial and structural details, while deeper layers encode high-level semantic representations. These hierarchical features exhibit strong interdependencies, making cross-level feature fusion crucial. To address this, MAIM is applied at every feature level to capture temporal feature variations across different hierarchical layers. This ensures that fine details, such as weak-information targets, are preserved while effectively incorporating relevant contextual and semantic information. As a result, the fusion process enhances the model's ability to distinguish changes accurately, improving the overall detection performance.

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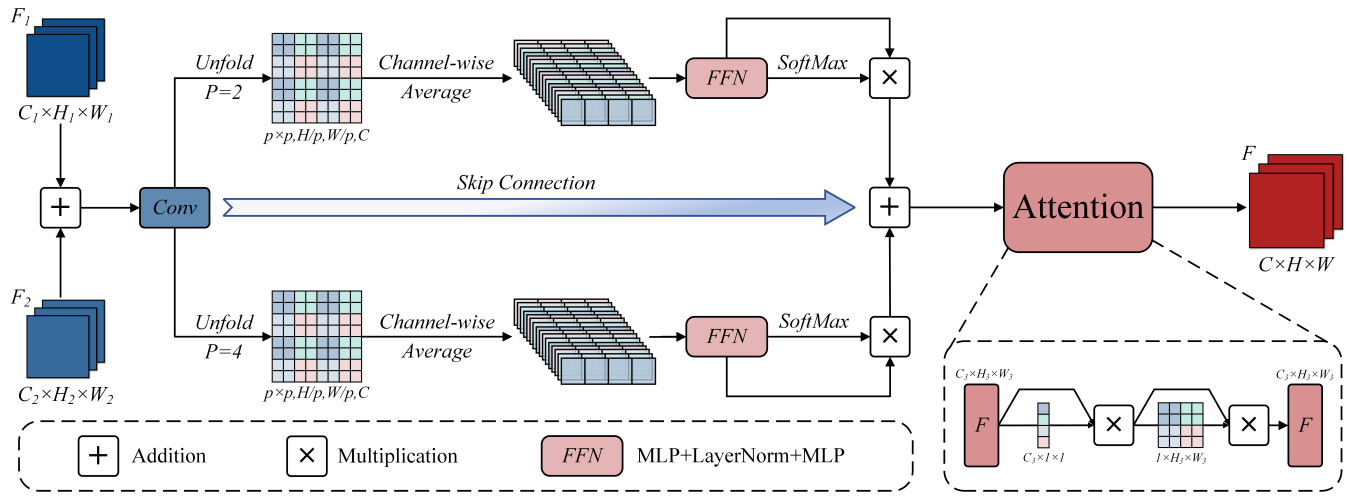


Fig. 4. Proposed MAIM. Aggregates local–global features via nonoverlapping patches and channel–spatial attention, preserving weak-target continuity.

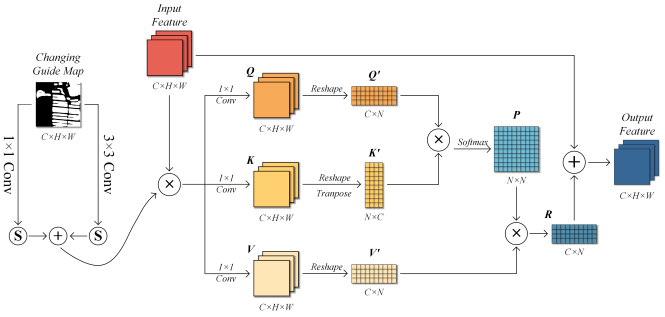


Fig. 5. Proposed CGA module. Dynamically weights features using a change map and multiscale convolution, with learnable intensity control (γ).

C. CGA Module

Against the backdrop of prioritizing prior knowledge generation, we introduce the CGA module, which leverages guidance mapping to enhance the network's ability to capture change information. This represents a novel feature fusion strategy that directs the network's focus toward size-driven changes in weak-information geographic features. Specifically, the architecture of the CGA module is illustrated in Fig. 5.

During the initialization of the CGA module, we define query, key, and value convolutional layers, all utilizing 1×1 convolutional kernels to extract feature information along the depth dimension. To achieve scene-adaptive optimization of feature responses, we introduce a learnable scaling parameter γ, which dynamically adjusts the intensity of the attention mechanism. This ensures that in weak-information target detection, the attention mechanism balances intensity-avoiding excessive noise introduction while preventing the critical information loss.

During the forward propagation of the CGA module, we use the change-guidance map G to guide the attention mechanism. It is important to clarify that G does not directly originate from the HFCM output feature F [see (5)]. Instead, it is obtained by concatenating the outputs of the bitemporal HFCM modules, performing multiscale feature fusion through the MAIM module, and finally generating a change-probability map via a lightweight decoder

$$G = \text{Sigmoid}\left(\text{Conv}_{1\times1}\left(\text{MAIM}\left(\text{Concat}\left(F_1, F_2\right)\right)\right)\right) \tag{11}$$

where F_1 and F_2 represent the outputs after dual-temporal encoding processed by the HFCM module.

Therefore, G is a single-channel probability map that encodes prior knowledge of potential change regions predicted from the fusion of bitemporal high-level semantic features, with a particular focus on indicating weak-information targets. In the CGA module, the guidance map G is first bilinearly upsampled to match the spatial dimensions ($H \times W$) of the input feature map F to be weighted. Subsequently, convolutional kernels of different scales are applied to G to extract multiscale contextual information

$$G_{\text{low}} = \text{Sigmoid}\left(\text{Conv}_{1\times1}\left(G\right)\right) \tag{12}$$

$$G_{\text{high}} = \text{Sigmoid}\left(\text{Conv}_{3\times3}\left(G\right)\right) \tag{13}$$

where G_{low} and G_{high}, respectively, capture the global and local contextual information within the prior change knowledge represented as G.

The two outputs are summed to form a multiscale guidance map $G_{\text{all}} = G_{low} + G_{\text{high}}$, which is elementwise multiplied with the input F, enhancing the feature representation. The resulting guidance-enhanced feature is then passed through three parallel 1×1 convolutions to generate the query (Q), key (K), and value (V) feature maps, each with shape $C \times H \times W$

$$Q = \text{Conv}\left(F\right), K = \text{Conv}\left(F\right), V = \text{Conv}\left(F\right). \tag{14}$$

These feature maps are reshaped: Q and V become $C \times N$, and K is reshaped and transposed to $N \times C$, where $N = H \times W$. An attention score matrix is then computed by

$$A = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \odot \left(G_{\text{low}} + G_{\text{high}}\right) \tag{15}$$

where d_k denotes the dimensionality of key vectors in attention mechanisms, used for scaling dot products to prevent gradient

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vanishing, $(d)_k^{1/2}$ is the scaling factor for feature dimension normalization and $\odot$ signifies elementwise multiplication, which is used to weight the value matrix via

$$R = A \cdot V \tag{16}$$

yielding an attention-enhanced feature of shape $C \times N$, which is reshaped back to $C \times N \times W$. A learnable scalar γ is applied to control the contribution of attention, and the final output is produced by residual addition

$$F_{\text{attn}} = \gamma \cdot (A \cdot V) + F \tag{17}$$

where $\cdot$ denotes the elementwise multiplication and γ is a learnable weight parameter.

This CGA module fuses multiscale guidance and spatial–channel attention, helping the model focus on weak-information targets even in small regions and suppress irrelevant background, ultimately improving detection accuracy.

D. Semantic Change Map Generation

The final output of SGI-Net is a semantic change map, which precisely indicates the change status at each pixel location in the image. For unchanged regions, the map retains the predicted semantic class; For changed regions, it explicitly encodes the "from–to" semantic transition (e.g., "building → bare land" and "water → farmland"). This semantic change map is generated by integrating the outputs from the network's two branches.

CD Branch: Outputs a single-channel binary change map $P_{cd} \in [0,1]^{H \times W}$, where each pixel value represents the probability of change between the bitemporal images. A threshold of 0.5 is used to determine change status: $P_{cd}(i,j) \geq 0.5$ indicates change, while values below the threshold indicate no change. This map is generated by a binary classifier applied to the fused features, consisting of a 1×1 convolutional layer followed by a Sigmoid activation function.

SS Branch: Outputs bitemporal SS maps. SS_1 Classifier: Predicts pixelwise semantic labels for time point T_1, producing a probability distribution $P_{ss_1} \in \mathbb{R}^{C \times H \times W}$, where C is the number of semantic classes (including the "no change" class). The final segmentation map $S_{t1} \in \{0,1,\ldots,C-1\}^{H \times W}$ is obtained via an argmax operation over the class dimension. SS_2 Classifier: Predicts pixelwise semantic labels for time point T_2, producing $P_{ss_2} \in \mathbb{R}^{C \times H \times W}$. Similarly, the segmentation map $S_{t2} \in \{0,1,\ldots,C-1\}^{H \times W}$ is obtained via argmax.

The semantic change map is generated through pixelwise logical inference: If $P_{cd}(i,j) < 0.5$, the pixel is labeled as "no change." If $P_{cd}(i,j) \geq 0.5$, a semantic transition pair $(C_{\text{from}}, C_{to})$ is assigned based on the predicted semantic labels $S_{t_1}(i,j)$ and $S_{t_2}(i,j)$. For example, if $S_{t_1}(i,j)$ ="building" and $S_{t_2}(i,j)$ ="bare land," the pixel is labeled as "building → bare land." This process is formally defined as follows:

$$\text{SCM}(i,j) = \begin{cases} \left(S_{t_1}(i,j), S_{t_2}(i,j)\right), & \text{if } P_{cd} \geq 0.5 \\ \text{Unchanged}, & \text{otherwise.} \end{cases} \tag{18}$$

E. Loss Function

For the SCD task, we use the binary cross-entropy (BCE) [57] loss function to compute the loss for the BCD results in the CD branch. In addition, we employ the categorical cross-entropy (CCE) [58], [59] loss function to calculate the final SCD results, specifically the losses for SCD1 and SCD2 [3]

$$L_{\text{total}} = L_{\text{BCE}} + \frac{1}{2}\left(L_{\text{Seg1}} + L_{\text{Seg2}}\right) \tag{19}$$

$$L_{\text{BCE}} = \frac{1}{N}\sum_i -\left[y_i \times log\,(p_i) + (1-y_i) \times \log\,(1-p_i)\right] \tag{20}$$

$$L_{\text{Seg}} = -\frac{1}{N}\sum_i \sum_{c=1}^{M} \log\,(p_{ic}). \tag{21}$$

Here, N represents the number of samples and y_i denotes the binary classification labels for CD, where changed categories are labeled as 1 and unchanged categories as 0. The corresponding p_i represents the predicted probabilities. In L_{Seg}, M represents the number of classes—for example, in the SECOND dataset, there are a total of seven classes, including the "no change" category. The variable y_{ic} denotes the ground-truth labels for SCD, while p_{ic} represents the predicted values for SCD [3].

III. Datasets Description and Experimental Settings

A. Datasets Environment

To demonstrate the model's effectiveness in SCD, this study will utilize the publicly available SECOND dataset along with our self-designed SJH_SCD dataset as the experimental basis to validate the proposed method. Example images from the datasets are shown in Figs. 6 and 7.

1) Second (SCD Dataset): The SECOND dataset [6] comprises 4662 pairs of images, with 2968 pairs available for training. The images correspond to geographical locations in cities such as Hangzhou, Chengdu, and Shanghai, with resolutions ranging from 0.5 to 3 m and dimensions of 512×512 pixels. This dataset includes seven categories: six land cover categories—water, ground, low vegetation, trees, buildings, and playgrounds—as well as a "no change" category. Each pair consists of two images captured at different times, with each image accompanied by an annotation map indicating the regions of temporal change and the land cover classes within those areas. We partitioned this dataset into training and test sets in a 4:1 ratio.

2) SJH_SCD: The SJH_SCD dataset is a self-constructed SCD dataset comprising 1468 pairs of images. This dataset utilizes Gaofen series satellite imagery (GF-1/2/6) based on Chizhou in Anhui Province and its surrounding areas (30.15°N–30.75°N and 117.10°E–117.85°E), covering different seasons to capture dynamic changes on the Earth's surface. The images were preprocessed through radiometric calibration, atmospheric correction, and orthorectification (RMSE < 0.5 pixel), feature mixed resolutions of 1 and 2 m with dimensions of 512×512 pixels. The dataset includes ten classes: nine land cover categories—bare land, roads, buildings, farmland, muddy areas, aquatic vegetation, water bodies, grassland, and woodland—as well as a "no change" category. Annotation was performed via manual visual interpretation with interannotator Kappa ≥ 0.85. We partitioned the dataset into training and testing sets in a 4:1 ratio.

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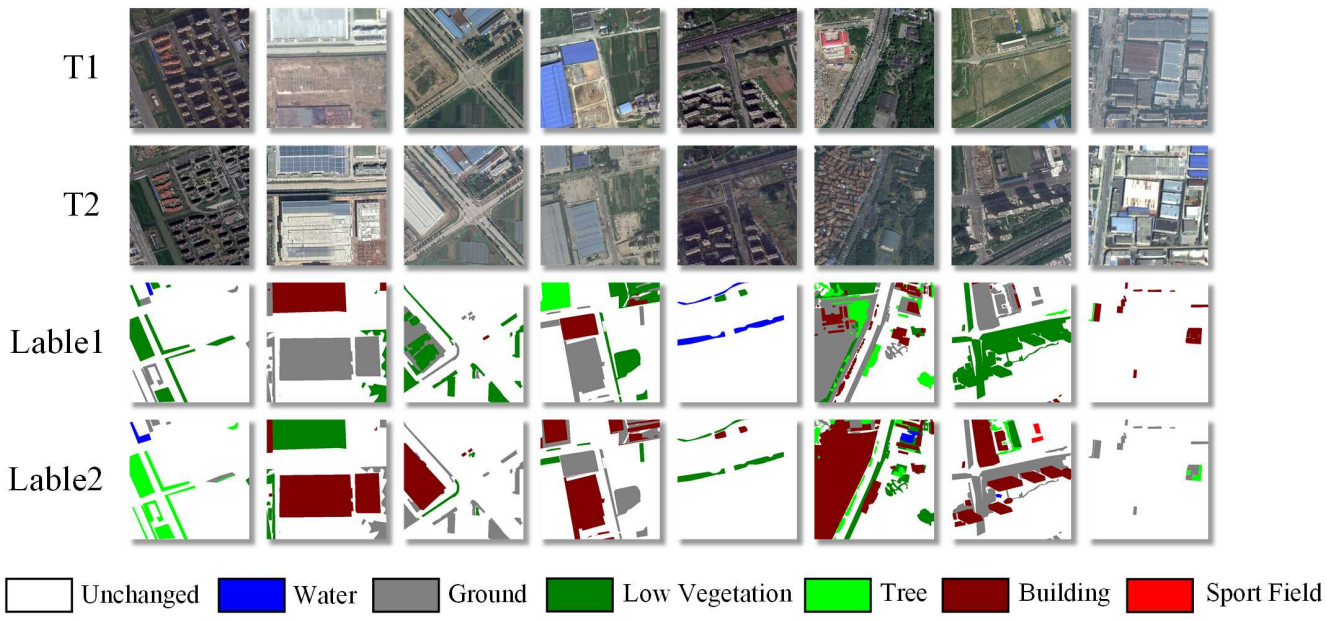


Fig. 6. Examples from the SECOND dataset. (a) T1 image, (b) T2 image, (c) change annotation, and (d) semantic annotation—categories: water (blue), ground (gray), low vegetation (light green), trees (dark green), buildings (red), and sports field (yellow).

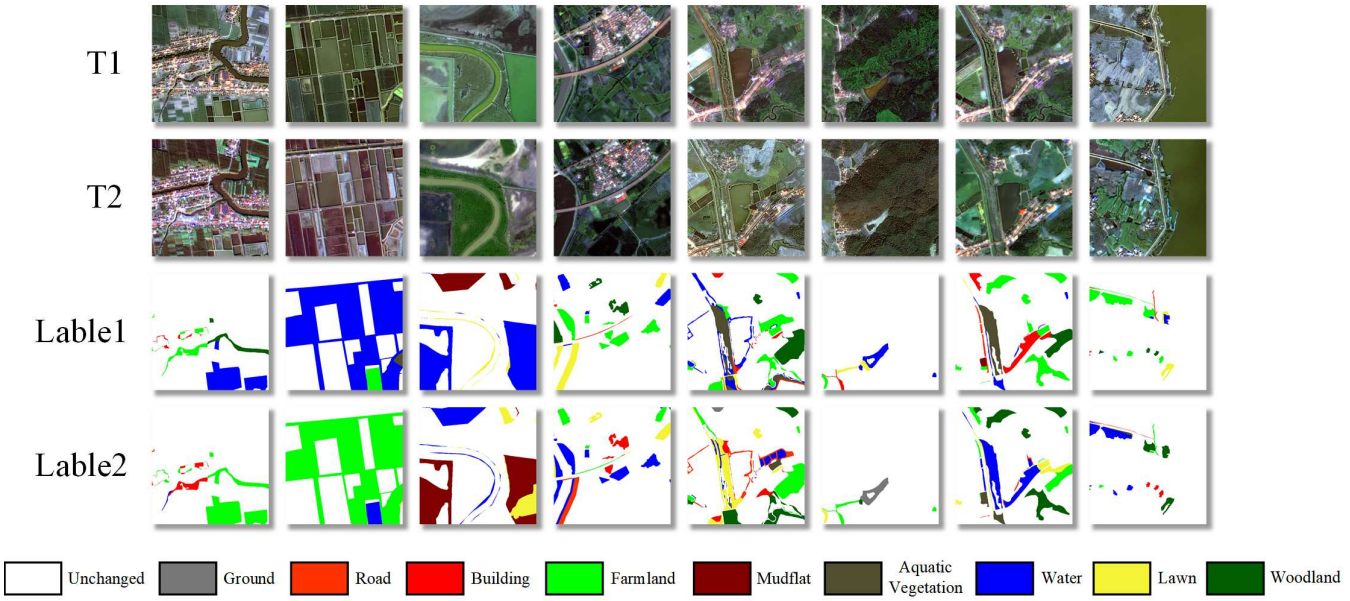


Fig. 7. Examples from the SJH_SCD dataset. (a) T1 image, (b) T2 image, (c) change annotation, (d) semantic map — categories: bare land (brown), road (black), buildings (red), farmland (orange), mudflat (purple), aquatic vegetation (cyan), water (blue), grassland (light green), and woodland (dark green).

B. Evaluation Metrics

For the SCD task, we evaluate our model's performance using the mean intersection over union (mIoU) [1], the separated Kappa coefficient (SeK) [3], the overall accuracy (OA) [27], and the per-image mIoU average. mIoU is the standard metric used in CD, while SeK is a metric introduced in ASN. We compute these evaluation metrics based on the confusion matrix generated from comparing the ground-truth labels with the predicted maps. The specific formulas are as follows:

$$\mathrm{IoU}_0 = \frac{\mathrm{TN}}{\mathrm{TN} + \mathrm{FP} + \mathrm{pn}} \tag{22}$$

$$\mathrm{IoU}_1 = \frac{\mathrm{TP}}{\mathrm{TN} + \mathrm{FP} + \mathrm{FN}} \tag{23}$$

$$\mathrm{mIoU} = (\mathrm{IoU}_0 + \mathrm{IoU}_1)/2. \tag{24}$$

Here, TP, FN, FP, and TN represent the number of true positives, false negatives, false positives, and true negatives at the pixel level, respectively, where pn denotes the number of false positives (FP) pixels in negative samples (i.e., predicted positives among actual negatives).

In addition, the predictions for each class of changed pixels are measured using SeK

$$\mathrm{kappa} = \frac{p_0 - p_e}{1 - p_e} \tag{25}$$

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$$\text{SeK} = e^{\text{IoU}_1 - 1} * \text{Kappa}. \tag{26}$$

Here, p_0 is defined as the sum of correctly classified samples across categories, divided by the total number of samples in each category. Meanwhile, p_e represents the cumulative product of the actual and predicted counts for all corresponding classes, which is then divided by the square of the total number of samples. The SeK metric is used to evaluate change patterns; however, due to the label imbalance in SCD, pixels that are truly 0 and predicted as 0 are excluded from the SeK computation and are assigned a value of 0. $Q = \{q_{ij}\}$ is the confusion matrix, where the element at position (i, j) represents the number of pixels belonging to class i that are classified as class j. The OA measures the proportion of correctly classified samples among all samples

$$OA = \sum_{i=0}^{N} q_{ii} \Big/ \sum_{i=0}^{N} \sum_{j=0}^{N} q_{ij}. \tag{27}$$

The per-image mIoU average is introduced as a supplementary metric to better evaluate detection performance, particularly for weak-information targets (e.g., small buildings and narrow roads) that may be overshadowed by dominant scene categories in conventional global metrics. It computes the binary IoU (change/nonchange) for each testing image independently using (23) and then averages these values across the entire testing set (N_{test}) images

$$PI_\text{mIoU} = \frac{1}{N_{\text{test}}} \sum_{k=1}^{N_{\text{test}}} \text{mIoU}^{(k)} \tag{28}$$

where $\text{mIoU}^{(k)}$ is the mIoU computed solely for the kth test image. This approach ensures equal weighting of all images regardless of their pixel distribution, making it more sensitive to improvements in detecting challenging weak targets and changes in small regions.

The combination of mIoU, SeK, and OA provides a more comprehensive evaluation of multiclass task performance. In addition, we introduce two metrics focused on assessing computational cost: the number of parameters (Params) [1], [4] and the number of floating-point operations (FLOPs) [1], [4]. To measure FLOPs, we consider the computation required for a pair of 512×512 pixel input images.

IV. Experimental Results and Discussion

In this section, we conduct a series of experiments to validate the effectiveness of the proposed SGI-Net architecture and its key components, namely, HFCM, MAIM, and CGA. Subsequently, we compare the proposed approach with several state-of-the-art (SOTA) methods recently employed in SCD.

The experiments in this study were conducted on a Windows 10 operating system, utilizing a 13th Gen Intel[1] Core[2] i7-13700F CPU and an NVIDIA GeForce RTX 4060 Ti GPU with 16-GB RAM. The models were implemented using the PyTorch framework. All experiments were performed under the same experimental settings, including a batch size of 8, 50 training epochs, an initial learning rate of 0.1, and a weight decay of 0.0005. The optimization was carried out using stochastic gradient descent (SGD) with Nesterov momentum [60]. Image augmentation strategies included random flipping and rotation of image pairs during training [1], [4].

[1]Registered trademark.
[2]Trademarked.

A. Comprehensive Comparison

To comprehensively evaluate the performance of the proposed SGI-Net, we compare it with several SOTA methods through SCD experiments. The compared methods include the following.

1) *HRSCD3 and HRSCD4 [27]:* HRSCD is a model architecture specifically designed for the SCD task and represents a pioneering study in applying deep learning to SCD. It provides two approaches: HRSCD3 and HRSCD4. HRSCD3 employs a two-stage training system, where two independent networks separately handle SS and BCD tasks. In contrast, HRSCD4 adopts a three-branch encoder structure, incorporating residual modules and an encoder–decoder architecture to enhance the detection of semantic changes over time while leveraging these components to improve the overall model performance.
2) *SSCDl and BiSRNet [3]:* SSCDl and BiSRNet are two models designed for CD. SSCDl utilizes a Siamese CNN encoder to extract semantic features, which are then fed into a CD decoder to effectively capture differences between images. Building upon SSCDl, BiSRNet further introduces global self-attention (SR) and cross-temporal SR (Cot-SR) modules. These enhancements strengthen the interaction between the temporal branch and the CD branch, improving the model's ability to process temporal and change-related features.
3) *TBFFNet [1]:* TBFFNet is a deep learning architecture designed for SCD in RSIs. Its specially designed CD Branch computes feature differences between two temporal phases and integrates multilevel features to generate change maps. The dual-stream design of TBFFNet enables it to simultaneously capture temporal changes and spatial features, improving the accuracy of CD. In addition, the use of upsampling operations ensures that the output results are aligned with the original input image dimensions.
4) *SCanNet [5]:* SCanNet is a specialized semantic change Transformer (SCanFormer) designed to model the "from–to" semantic transitions in dual-temporal RSIs. It utilizes a variant of the cross-shaped window Transformer, which is more effective in capturing semantic changes in dual-temporal RSIs.
5) *DFINet [4]:* DFINet is an innovative SCD network that focuses on mining both intraclass and interclass differences in dual-temporal RSIs. The network adopts a dual-dimensional feature interaction strategy, aiming to address the limitations of existing methods in temporal feature extraction. Its advantage lies in its ability to not only focus on spatial features but also fully consider temporal features, effectively reducing the false change phenomena and improving the accuracy of SCD.

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TABLE I
QUANTITATIVE RESULTS ON THE SECOND DATASET AND SJH_SCD DATASET

Methods	SECOND					SJH_SCD					Flops(G)	Params(Mb)
	mIoU(%)	SeK(%)	OA(%)	Per-Image mIoU(%)	Predict Time(s)	mIoU(%)	SeK(%)	OA(%)	Per-Image mIoU(%)	Predict Time(s)		
HRSCD3	66.23	12.07	84.64	62.85	72.86	64.82	9.74	84.08	52.44	**33.60**	42.94	**12.78**
HRSCD4	71.17	19.13	85.84	68.42	73.01	73.52	23.32	87.77	59.36	33.80	43.97	13.71
SSCDl	72.88	22.33	87.32	70.36	**72.11**	75.71	28.82	89.02	62.29	34.10	**35.47**	23.31
BiSRNet	72.79	22.00	87.36	70.10	72.83	75.05	27.75	88.37	61.73	34.39	190.30	23.38
TBFFNet	73.13	21.85	87.29	70.47	90.46	75.52	27.48	88.08	63.11	45.25	401.09	26.02
SCanNet	72.49	21.56	87.48	69.66	81.71	75.27	29.46	88.78	62.15	40.16	258.74	27.90
DFINet	73.27	23.01	87.50	70.65	73.92	75.40	30.21	88.87	62.49	36.67	204.06	24.26
SGI-Net(Ours)	**73.86**	**23.57**	**87.84**	**71.17**	72.56	**77.43**	**33.10**	**89.65**	**64.80**	36.44	189.56	24.33

The quantitative experimental results (see Table I) demonstrate the performance improvements of SGI-Net across different datasets, with the highest scores highlighted in bold. On the SECOND dataset, compared with the latest TBFFNet, SCanNet, and DFINet networks, SGI-Net has improved the mIoU by 0.73%, 1.37%, and 0.59%, respectively, the SeK by 1.72%, 2.01%, and 0.56%, respectively, and the OA by 0.55%, 0.36%, and 0.34%, respectively. On the SJH_SCD dataset, SGI-Net has improved the mIoU by 1.91%, 2.16%, and 2.03%, respectively, the SeK by 5.62%, 3.64%, and 2.89%, respectively, and the OA by 1.57%, 0.87%, and 0.78%, respectively. These results indicate that SGI-Net exhibits a certain level of generalization ability across different datasets, particularly in the SeK metric, further confirming its accuracy and stability in change region identification.

To comprehensively evaluate the detection performance of weak-information targets (e.g., small buildings and narrow roads) that may be overshadowed by dominant scene categories in conventional global metrics, we introduce the per-image mIoU average as a supplementary evaluation criterion. This approach computes the binary IoU (change/nonchange) for each testing image independently before averaging across the entire dataset, ensuring equal weighting of all images regardless of their pixel distribution. As demonstrated in Table I, SGI-Net achieves SOTA per-image mIoU scores of 71.17% on SECOND and 64.80% on SJH_SCD, outperforming the strongest baseline TBFFNet by 0.70% and 1.69%, respectively. The higher sensitivity of this metric to challenging weak targets stems from its avoidance of dominance by large foreground/background areas. The consistent superiority across both global and per-image metrics validates our method's robustness, with these gains directly attributable to: 1) HFCM's preservation of fine-grained details through dual-path decoupling; 2) MAIM's maintenance of structural continuity; and 3) CGA's dynamic enhancement of subtle change signals. The 72.56-s prediction time further confirms the practical efficiency of our approach in real-world applications.

To verify that the observed gains of SGI-Net over TBFFNet and DFINet on the SECOND dataset are not due to random-seed variability, we performed three independent runs (random seeds: 42, 123, and 2025) and report mean ± standard deviation for both mIoU and SeK. The descriptive statistics are summarized in Table II. Next, we conducted two-tailed paired t-tests using the results from three independent runs to determine whether the performance gains of SGI-Net over TBFFNet and DFINet are statistically significant at the 5% level. For the comparison with TBFFNet, the t-test yielded $t(2) = 44.80$ with $p = 0.00050$ for mIoU and $t(2) = 38.12$ with $p = 0.00069$ for SeK. Similarly, in the comparison with DFINet, the test yielded $t(2) = 38.77$ with $p = 0.00066$ for mIoU, and $t(2) = 5.12$ with $p = 0.03603$ for SeK. All p-values are below the 0.05 threshold, indicating that the improvements brought by SGI-Net are statistically significant across both metrics. Given the small sample size ($n = 3$), we also applied the nonparametric Wilcoxon signed-rank test. In each comparison, the test statistic was 0 (the smallest possible), corresponding to $p = 0.25$. As expected, the Wilcoxon test lacks power for $n = 3$; nonetheless, its zero-rank statistics indicate that SGI-Net's scores were uniformly higher across seeds. In summary, both parametric and nonparametric analyses support the reliability of SGI-Net's performance improvements over TBFFNet and DFINet on the SECOND dataset. These results, together with its consistent gains in OA (not shown here), demonstrate that the improvements are unlikely to be attributable to random-seed variance.

TABLE II
DESCRIPTIVE STATISTICS ON THE SECOND DATASET

Methods	mIoU(%)	SeK(%)
TBFFNet	73.10 ± 0.08	21.92 ± 0.12
DFINet	73.28 ± 0.08	22.74 ± 0.30
SGI-Net(Ours)	**73.84 ± 0.05**	**23.58 ± 0.04**

To more intuitively compare the computational performance of our model, as shown in Table III, we calculate the inference speed by dividing the total prediction time for the 593 testing

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Fig. 8. Comparison experiments on the SECOND dataset. The main differences are highlighted with orange rectangles.

TABLE III
QUANTITATIVE COMPARISON OF COMPUTATIONAL EFFICIENCY ON THE SECOND DATASET

Methods	Flops(G)	Params(Mb)	Time(s)	mIoU(%)	SeK(%)
BiSRNet	190.30	**23.38**	**0.168**	72.79	22.00
TBFFNet	401.09	26.02	0.235	73.13	21.85
SCanNet	258.74	27.90	0.188	72.49	21.56
DFINet	204.06	24.26	0.171	73.27	23.01
SGI-Net(Ours)	**189.56**	24.33	0.186	**73.86**	**23.57**

images in the SECOND dataset by the number of images, i.e., Inference Time (S) = Total Prediction Time/Number of Test Images. From the table, we can observe that SGI-Net achieves floating-point operations per second (FLOPs) (189.56G) lower than many recent mainstream methods, even outperforming the lightweight model BiSRNet. The number of parameters of SGI-Net (24.33 Mb) is nearly the same as DFINet (24.26 Mb), but SGI-Net achieves a 0.59% improvement in mIoU and a 0.56% increase in SeK. SGI-Net's inference time (0.186 s) is comparable to BiSRNet (0.168 s) and DFINet (0.171 s), and is significantly faster than TBFFNet (0.235 s). Although the current computational load has not yet reached an ultralightweight level, we have already adopted the following optimization strategies: DSCs are used in the HFCM and MAIM modules to reduce computational complexity. The MAIM module adaptively adjusts the patch size based on input complexity (e.g., using larger patches for simpler backgrounds), which helps to reduce the inference time.

Figs. 8 and 9 present the visual detection results of SGI-Net on different datasets. Fig. 8 focuses on the detection of fine-grained object changes, such as buildings and water bodies. In the pair of images (1) and (2), where small-scale building changes are present, SGI-Net accurately identifies the target areas, significantly reducing the occurrence of missed detections compared with other models. This demonstrates the higher sensitivity of the proposed method in detecting building changes in complex terrains. In the pair of images (4), where small object changes are numerous and complexly distributed, SGI-Net still maintains high responsiveness to small building changes and performs excellently in contour and boundary refinement. In the pair of images (3), SGI-Net not only accurately detects overall changes but also identifies small water body changes with precision, showing a clear advantage

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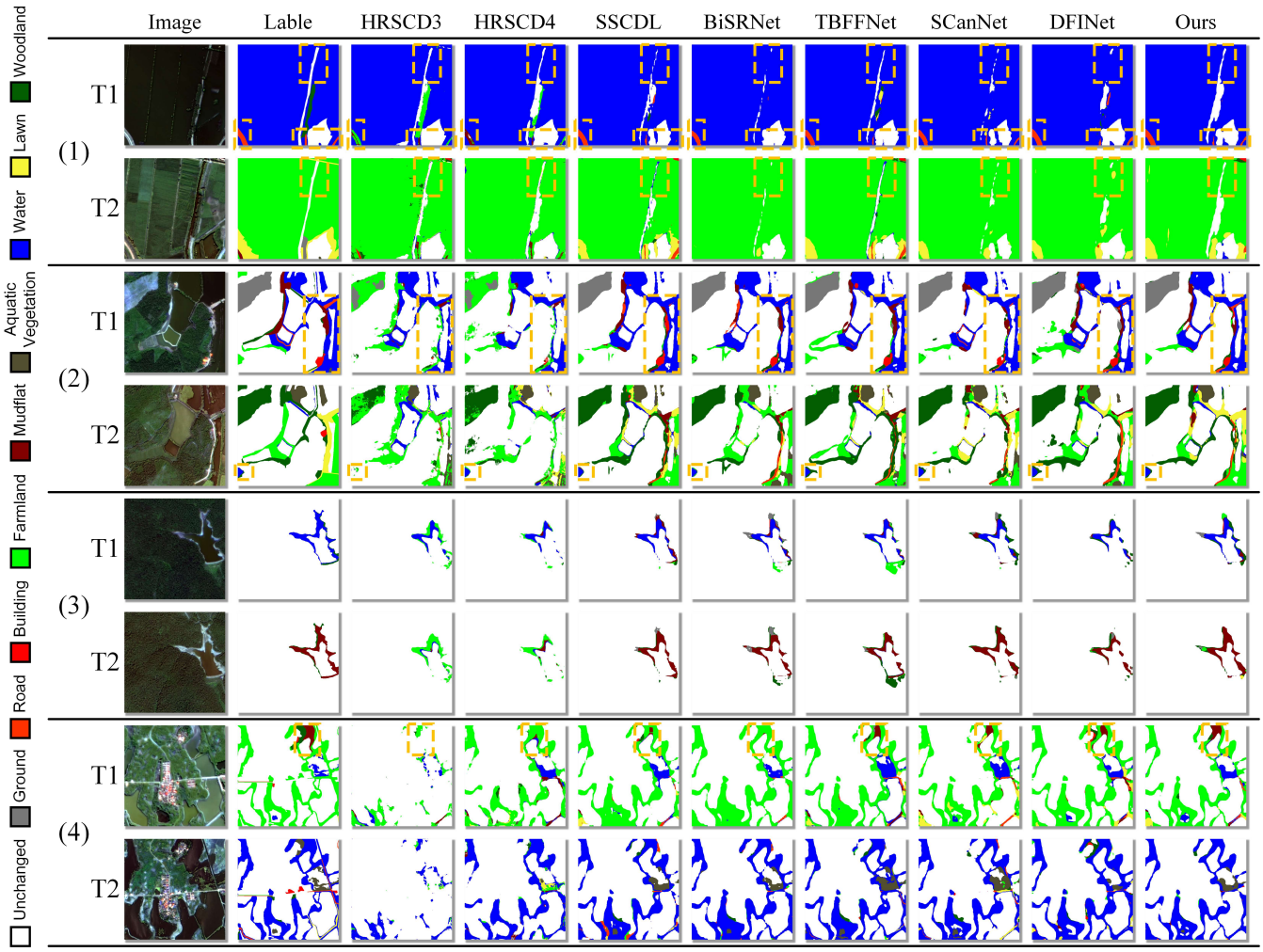


Fig. 9. Comparison experiments on the SJH_SCD dataset. The main differences are highlighted with orange rectangles.

in boundary integrity compared with other models. Fig. 9 mainly addresses changes in features such as farmland, water bodies, and mudflats. In the pair of images (1), where large-scale water bodies change to farmland, SGI-Net is still able to detect small building changes accurately and outperforms other models in terms of small road integrity. In the pair of images (2), where the change background is complex with many types of features, the detection task is challenging. However, SGI-Net still captures small water body and building changes with high precision, maintaining a high sensitivity. In the pair of images (3) and (4), SGI-Net performs excellently in detecting small changes from water bodies to mudflats. While other models can detect some contour changes in the pair of images (3), they struggle to detect the finer mudflat-water transitions in the pair of images (4). SGI-Net, however, provides more complete and accurate detection results for both the subtle contour changes in the pair of images (3) and the feature transitions in the pair of images (4). This superiority is primarily attributed to the introduction of the HFCM module, which ensures that shallow fine-grained information is fully retained during the encoding phase, and through the MAIM module, multistage information interaction is achieved during feature fusion, enabling precise identification of small-scale targets and improving the overall detection accuracy.

B. Ablation Study

To demonstrate the effectiveness of the HFCM, MAIM, and CGA modules, ablation experiments will be conducted on the SECOND and SJH_SCD datasets.

As shown in Table IV, the CGA, MAIM, and HFCM modules improved the mIoU by 0.32%, 0.78%, and 0.91% on the SECOND dataset, and by 0.07%, 0.33%, and 0.70% on the SJH_SCD dataset, respectively. Specifically, HFCM significantly enhances fine-grained boundary detection through cross-level feature fusion and dual-path disentanglement. Compared with the baseline, HFCM alone improves mIoU by 0.25%, while reducing the number of parameters (23.22M versus 23.28M), demonstrating the efficiency of its lightweight design. MAIM employs a nonoverlapping patchwise strategy and dual-domain attention mechanism to effectively enhance contextual associations for weak-information targets. It improves mIoU by 0.42% with only a 1.9% increase in FLOPs (188.70G versus 185.18G), validating its computational efficiency. Notably, on the SJH_SCD dataset, MAIM

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TABLE IV
ABLATION RESULTS ON THE SECOND DATASET AND SJH_SCD DATASET

Methods	SECOND			SJH_SCD			Flops(G)	Params(Mb)
	mIoU(%)	SeK(%)	OA(%)	mIoU(%)	SeK(%)	OA(%)		
Base	72.95	22.16	87.34	76.73	32.03	89.56	**185.18**	23.28
Base+HFCM	73.20	22.57	87.56	76.90	32.07	89.30	185.48	**23.22**
Base+MAIM	73.37	22.93	87.62	77.16	32.22	89.28	188.70	24.22
Base+CGA	73.37	22.94	87.74	76.80	32.03	89.38	188.63	24.11
Base+HFCM+MAIM	73.40	22.97	87.44	77.29	32.94	89.48	189.00	24.26
Base+HFCM+CGA	73.28	22.68	87.61	77.21	32.41	89.47	188.93	24.15
Base+MAIM+CGA	73.73	23.38	87.62	77.06	32.60	89.57	189.26	24.29
SGI-Net(Ours)	**73.86**	**23.57**	**87.84**	**77.43**	**33.10**	**89.65**	189.56	24.33

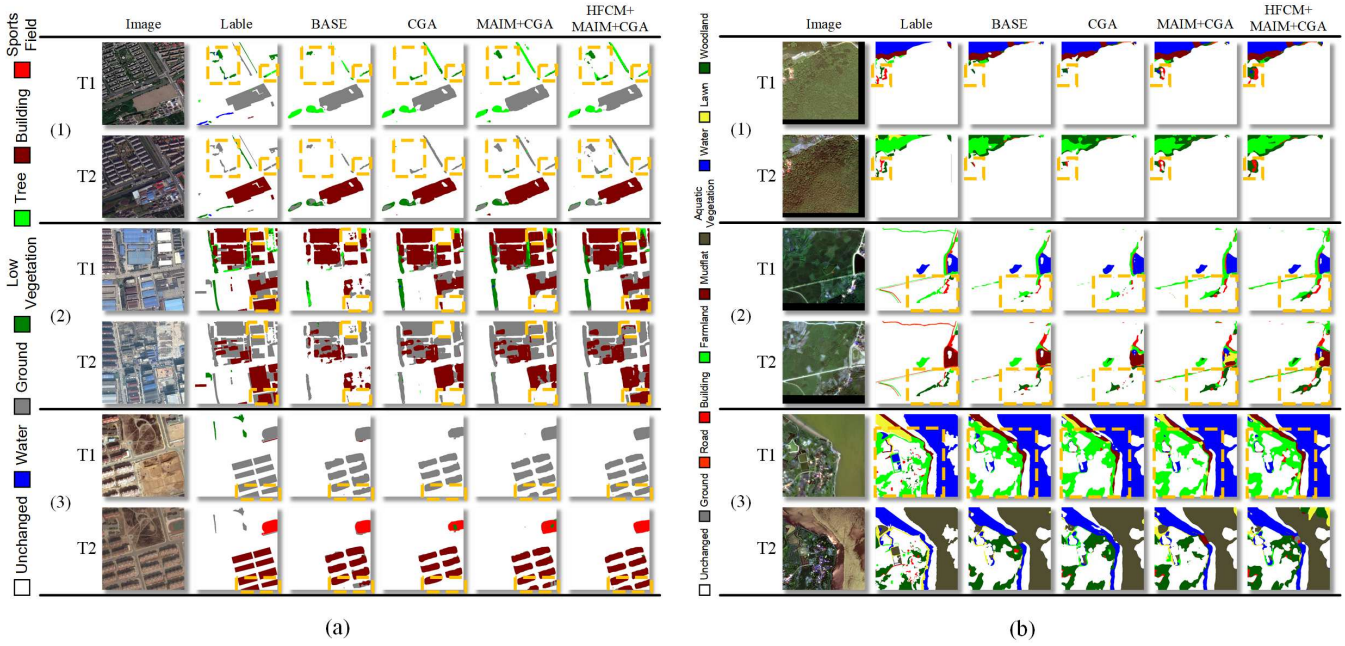


Fig. 10. Qualitative results of the ablation study for the CGA, MAIM, and HFCM modules on (a) SECOND dataset and (b) SJH_SCD dataset. The main differences are highlighted with orange rectangles.

achieves a +0.43% mIoU gain under mixed-resolution inputs (1 m/2 m), indicating that its patching strategy effectively preserves spatial continuity. The CGA module leverages change priors generated by HFCM to dynamically suppress background interference, resulting in a 0.42% mIoU improvement. The synergy between HFCM and MAIM allows HFCM to provide geometric priors while MAIM enhances local–global interaction, achieving a combined mIoU gain of 0.45%. When HFCM is paired with CGA, the integration of geometric priors and attention refinement leads to a 0.52% SeK improvement (22.68 versus 22.16). The full model, incorporating all three modules, yields the highest overall gain (mIoU +0.91% and SeK +1.41%), validating the effectiveness and rationality of the proposed architecture.

Visual comparison results are shown in Fig. 10, where the results after adding the CGA, MAIM, and HFCM modules are displayed from left to right. In the SECOND dataset, Fig. 10(a-1) indicates that after introducing these modules, the model effectively captures subtle changes in low vegetation and trees transitioning to bare land. In the more complex scene of Fig. 10(a-2), the inclusion of MAIM and HFCM modules significantly improves the detection accuracy of dense building change areas, preventing missed detections. Moreover, Fig. 10(a-3) further validates that with the introduction of these modules, the model's performance in detecting fine-grained changes is notably enhanced, with improved boundary delineation compared with the baseline model. On the SJH_SCD dataset, Fig. 10(b-1) shows that the model's sensitivity to small building changes has been improved. Fig. 10(b-2) indicates that in a scenario where farmland transitions to roads, the addition of the MAIM and HFCM modules significantly enhances the integrity of target extraction. Additionally, in

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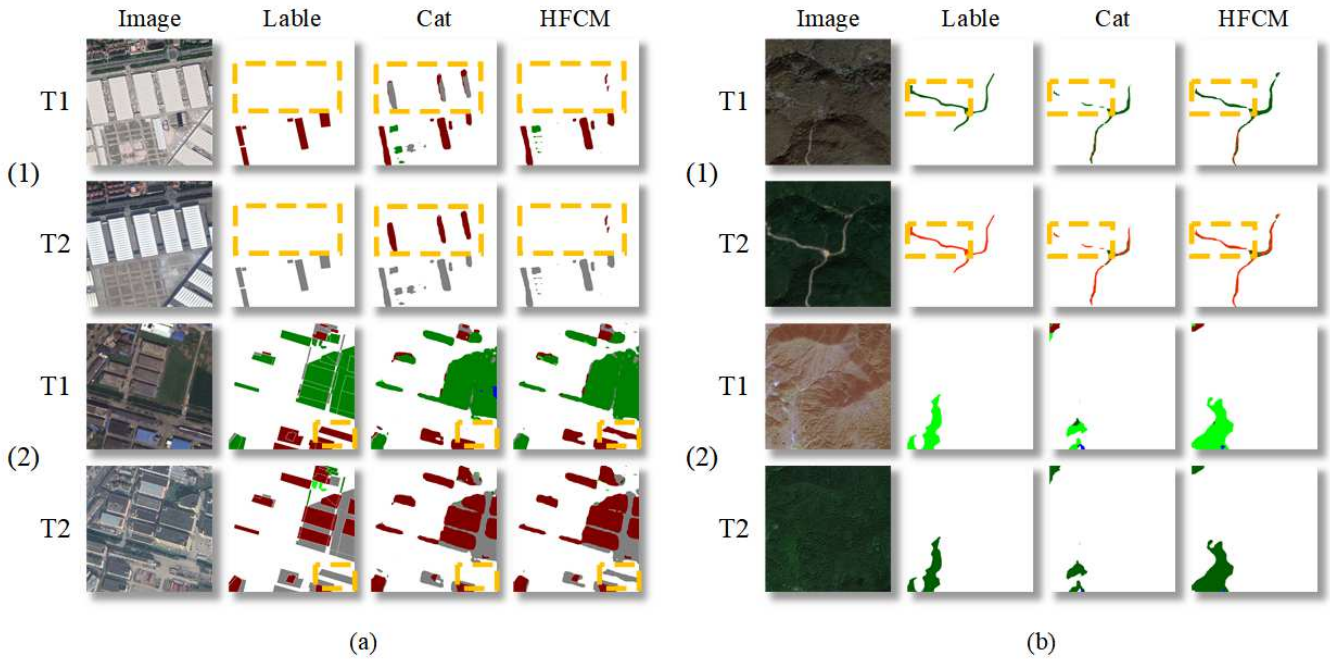


Fig. 11. (a) and (b) Comparison results of different feature extraction methods on the SECOND dataset and the SJH_SCD dataset, respectively.

TABLE V
COMPARISON RESULTS OF DIFFERENT FEATURE EXTRACTION METHODS ON THE SECOND DATASET AND SJH_SCD DATASET

Methods	SECOND				SJH_SCD				Flops(G)	Params(Mb)
	mIoU(%)	SeK(%)	OA(%)	Per-Epoch/min	mIoU(%)	SeK(%)	OA(%)	Per-Epoch/min		
Cat	73.19	22.66	87.25	7.71	76.93	31.5 6	89.26	7.84	**189.53**	**24.32**
HFCM	**73.86**	**23.57**	**87.84**	**7.48**	**77.43**	**33.10**	**89.65**	**4.29**	189.56	24.33

TABLE VI
COMPARISON RESULTS OF DIFFERENT FEATURE FUSION METHODS ON THE SECOND DATASET AND SJH_SCD DATASET

Methods	SECOND				SJH_SCD				Flops(G)	Params(Mb)
	mIoU(%)	SeK(%)	OA(%)	Per-Epoch/min	mIoU(%)	SeK(%)	OA(%)	Per-Epoch/min		
Sub	73.09	22.37	87.34	8.26	76.88	32.48 6	89.63	**3.94**	**188.93**	**24.15**
MAIM	**73.86**	**23.57**	**87.84**	**7.48**	**77.43**	**33.10**	**89.65**	4.29	189.56	24.33

the case of large-scale complex changes in Fig. 10(b-3), the proposed modules not only ensure the accuracy of large-scale CD but also enhance sensitivity to subtle changes, allowing the model to detect fine changes that were previously undetectable without the modules. This result fully demonstrates the effectiveness of the proposed modules in weak-information targets extraction and improves the overall detection accuracy.

C. Feature Fusion Strategies Analysis

To further validate the effectiveness of the proposed HFCM and MAIM modules, we conducted comparative experiments with traditional feature extraction and fusion methods. The quantitative comparison results of different feature extraction strategies are shown in Table V. On both datasets, the HFCM module shows a clear advantage over the direct channel concatenation approach for shallow and deep layer features, as it effectively merges deep semantic information and shallow detail features while refining the extracted features. On the SECOND and SJH_SCD datasets, HFCM improved mIoU by 0.67% and 0.50%, SeK by 0.91% and 1.54%, and OA by 0.59% and 0.39%, respectively. Moreover, with almost the same computational cost, HFCM achieved faster inference speeds, offering better detection efficiency compared with traditional methods. The corresponding qualitative results are shown in Fig. 11. For the MAIM module, we further compared it with traditional dual-temporal feature fusion methods. The corresponding quantitative results are shown in Table VI. The experimental results demonstrate that the proposed multiscale parallel fusion strategy improves accuracy compared with

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TABLE VII
COMPARISON RESULTS OF DIFFERENT HIERARCHICAL FEATURE INPUTS FOR HFCM ON THE SECOND DATASET AND SJH_SCD DATASET

HFCM Input Layers	SECOND				SJH_SCD				Flops(G)	Params(Mb)
	mIoU(%)	SeK(%)	OA(%)	Per-Epoch/min	mIoU(%)	SeK(%)	OA(%)	Per-Epoch/min		
L0	73.43	22.84	87.48	8.26	77.05	32.48	89.39	4.28	191.68	**24.31**
L1	73.27	22.80	87.33	8.04	76.47	31.35	89.26	4.41	189.87	**24.31**
L4	73.38	22.81	87.55	8.11	76.93	32.41	**89.72**	**4.00**	**189.56**	24.33
L0+L1	73.39	22.87	87.40	7.97	77.18	32.54	89.27	4.52	**189.56**	24.33
L0+L4	73.55	23.09	87.60	7.54	76.80	31.84	89.52	4.02	190.61	24.34
L1+L4	73.69	23.22	87.51	7.74	76.89	32.21	89.58	4.06	189.83	24.34
L0+L1+L4	**73.86**	**23.57**	**87.84**	**7.48**	**77.43**	**33.10**	89.65	4.29	**189.56**	24.33

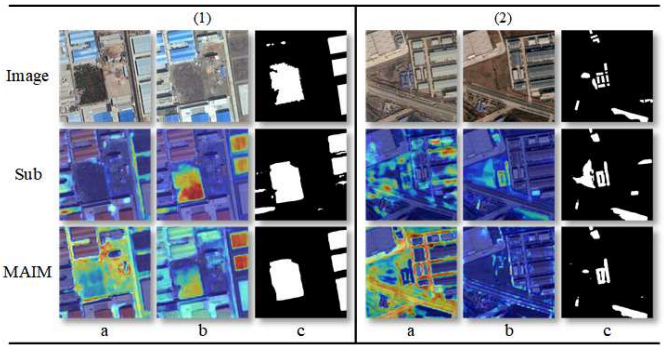


Fig. 12. Visualizationresults of heatmaps on the SECOND dataset under different fusion methods. From top to bottom, (a) and (b) images from different periods along with the corresponding heatmaps generated by various methods, and (c) labels and predicted values.

traditional image differencing methods. Specifically, on the SECOND and SJH_SCD datasets, MAIM improved mIoU by 0.77% and 0.55%, SeK by 1.20% and 0.62%, and OA by 0.10% and 0.02%, respectively.

In addition, to further explore the impact of different feature fusion methods on the model's focus areas, we generated visual heatmaps of different feature fusion methods on the SECOND dataset using gradient-weighted class activation mapping (Grad-CAM) [60], as shown in Fig. 12. The results demonstrate that the proposed method shows more stable performance when handling the same types of features. Furthermore, by comparing the heatmap changes of different models during training, it can be observed that the MAIM module more effectively focuses on fine-grained details of weak-information targets such as small landform features. Compared with traditional feature fusion methods, MAIM better retains detailed information during the dual-temporal image processing, leading to an improvement in overall detection accuracy.

D. Hierarchical Feature Selection in HFCM

To further validate the design of the HFCM module proposed in Section II—which leverages two layers of shallow features (Layer0 and Layer1) in conjunction with the deepest semantic feature (Layer4) to produce representative prior knowledge for guiding subsequent feature fusion—we conducted a series of ablation experiments in which HFCM was driven by different subsets of ResNet34's feature layers. ResNet34 is divided into six stages: Layer0, Maxpool, Layer1, Layer2, Layer3, and Layer4. Layer0 (immediately following the input convolution) and the Maxpool stage extract low-level features at high spatial resolution, capturing fine details such as textures and edges; Layer1 and Layer2 extract mid-level features, gradually enriching semantic information while reducing spatial resolution; Layer3 and Layer4 extract high-level features rich in global semantics, which are essential for detecting changes in complex scenes. From the perspective of feature-pyramid theory and the information-bottleneck principle, these stages exhibit clear hierarchical dependencies: shallow layers retain pixel- and object-level geometric details that are indispensable for localization, deep layers compress redundant low-level signals to emphasize task-relevant semantics, and mid-level layers serve as a structured bridge that correlates fine details with global context. Selecting Layer0, Layer1, and Layer4 thus provides minimal redundancy coverage across abstractions—from pixel-level to scene-level—while avoiding overlap in Layer2/Layer3 and maintaining an optimal computation–accuracy tradeoff.

Table VII reports the quantitative results for inputting various combinations of these feature layers into HFCM on the SECOND dataset. When comparing HFCM configurations that fuse multiple feature layers versus single-layer inputs, we observe no increase in total FLOPs—for instance, fusing L0, L1, and L4 yields the same network FLOPs (189.56G) as fusing only L0 and L1—yet, the single-epoch training time decreases modestly while the overall parameter count increases only slightly (e.g., 23.33M versus 23.28M). This indicates that the inclusion of deeper semantics (L4) does not introduce extra computational burden but does raise parameter capacity in a way that accelerates convergence. Adding L4 on top of shallow features L0 and L1 increases mIoU by 0.47% and SeK by 0.70%, confirming that L4's global semantic context substantially enhances detection of weak-information targets.

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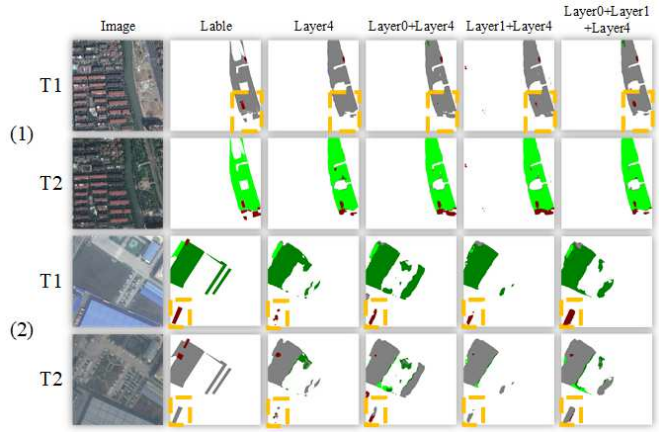


Fig. 13. Qualitative results of different hierarchical feature inputs for HFCM on the SECOND dataset.

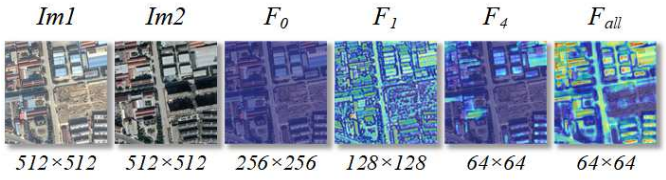


Fig. 14. Grad-CAM visualization results of multilevel features in the HFCM module on the SECOND dataset.

Each combination with L4 alone outperforms configurations that use only shallow features in terms of SeK, demonstrating that fusing L4 compensates for the spatial resolution limitations of any single shallow feature layer. Using L4 by itself yields a lower SeK (22.81%) compared with fusing all three layers (23.57%), indicating that although deep features provide strong semantics, they cannot fully capture geometric details without complementary shallow features. On mixed-resolution farmland scenes in the SJH_SCD dataset, the L0+L1+L4 configuration yields a +0.56% SeK gain, indicating that L4's global semantics help distinguish easily confused classes. Furthermore, the OA of L0+L1 (89.27%) is lower than that of L0+L1+L4 (89.65%), validating L4's role in improving global consistency.

Furthermore, the qualitative results in Fig. 13 further confirm that fusing multilevel information helps the model more accurately recognize small features, and outperforms single-feature input in terms of target extraction completeness. Meanwhile, as shown in the Grad-CAM heatmaps in Fig. 14, the analysis further reveals the underlying mechanism of multilevel feature fusion within the HFCM module: the shallow feature F_0 preserves abundant edge details but suffers from significant noise, whereas the deep feature F_4 exhibits strong semantic responses but loses fine structural information. Notably, the downsampled and fused feature F_{all}, through the dual-path mechanism, enhances the high-frequency residuals (e.g., contours of small buildings) retained from F_0/F_1 in the subtractive path, while integrating the semantic priors of F_4 in the additive path. As a result, it maintains the gradient response strength of weak targets even at a low resolution of 64×64. This finding is consistent with the complete detection results of small targets using L0+L1+L4, as shown in Fig. 13, indicating that the downsampling process does not simply discard information but instead distills and reorganizes key geometric cues through feature coupling.

TABLE VIII
COMPARISON OF DIFFERENT PATCH SIZES IN THE MAIM MODULE ON THE SECOND DATASET

Configuration	Patch Size(p)	mIoU (%)	SeK (%)	OA (%)	Flops (G)	Params (Mb)
Local Branch Only	p=2	73.59	23.11	87.75	189.46	**24.29**
Global Branch Only	p=4	73.32	22.80	87.31	**189.45**	**24.29**
Dual-Branch Combo	**p=2,4**	**73.86**	**23.57**	**87.84**	189.56	24.33

E. Adaptive Patch Design in MAIM

Meanwhile, to validate the rationale behind the patch size p selection in the MAIM module and its impact on performance, we conducted ablation experiments on the SECOND dataset for three configurations: single local branch ($p = 2$), single global branch ($p = 4$), and dual-branch combination ($p = 2, 4$). The results are shown in Table VIII. The single local branch effectively preserves fine details of small targets but exhibits slightly lower SeK and OA due to the lack of global context; the single global branch enhances overall consistency but weakens responses for weak-information targets; the dual-branch combination, by parallel extraction of high-frequency details and low-frequency semantics, achieves the best balance between the detail enhancement and global consistency, achieving optimal performance (mIoU 73.86%, SeK 23.57%, and OA 87.84%) with only marginal increases in FLOPs and parameters, thereby demonstrating the effectiveness and efficiency of the multiscale patch strategy.

V. CONCLUSION

In this study, the proposed SGI-Net systematically resolves weak-information target detection challenges through a coordinated optimization mechanism combining semantic-driven hierarchical feature coupling, spatiotemporal adaptive reasoning, and prior-guided attention enhancement. Specifically: 1) the HFCM eliminates semantic ambiguity by establishing cross-layer geometry-semantic correlations; 2) the MAIM preserves spatiotemporal continuity through nonoverlapping patch-based fusion; and 3) the CGA module dynamically amplifies faint change signals using prior knowledge. This technical pathway achieves remarkable performance on SECOND and SJH_SCD datasets (mIoU improvements of 0.73%–2.16% and SeK gains of 1.26%–5.62%), with visualization analyses confirming enhanced feature responses for weak targets in complex scenes and improved fine-grained boundary discrimination. Ablation studies validate each module's contribution to precise localization and false-positive suppression.

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Our evaluation has so far been limited to high-resolution RGB imagery, leaving cross-sensor and cross-modal generalization untested. Future work will focus on extending SGI-Net to multispectral, SAR–optical, and low-resolution data, and on exploring domain-adaptation and weakly supervised strategies to broaden its applicability.

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